

Transparent Multi-Hop Routing for Raft

Under Partial Network Partitions

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Partial partitions can interrupt Raft

Nodes may remain reachable through peers even when their direct links fail.

Consensus services need progress to remain available.

RQ1 Progress

When does forwarding restore stable commits?

RQ2 Latency

What delay does forwarding add to client-observed commits?

Goal: improve resilience while keeping the Raft protocol unchanged.

Raft and the routing gap

Raft commits after a majority stores an entry.
A five-node cluster needs three replicas.

Jensen et al. (2021)

Propose overlays for Raft

Nifty (2020)

Reroutes traffic in other systems

Omni-Paxos (2023)

Changes consensus for partial connectivity

Our gap: measure routing benefits and costs beneath unchanged Raft.

Controlled network experiments

900

planned trials

5 topologies

2 communication modes

3 link delays

30 repetitions

HashiCorp Raft in Mininet

Direct links vs. preconfigured forwarding

Link delay: 0.5, 5 or 20 ms

Heartbeat: 10 ms

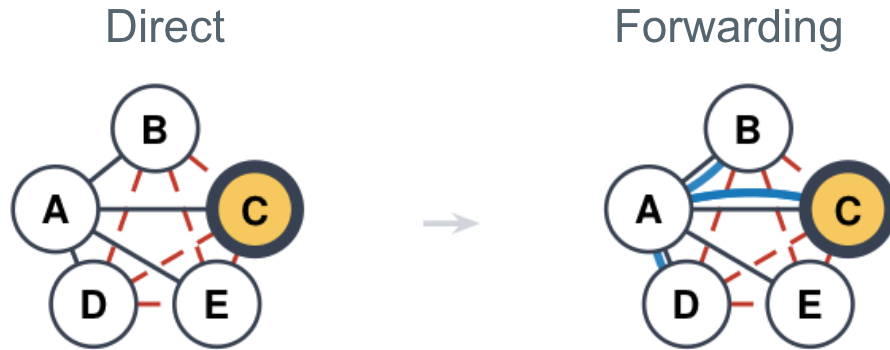
100 requests/s, 64-byte payloads

30 s warm-up, cut links, 30 s observation

607 valid runs. Primary comparisons use 0.5 ms links.

Partitions test quorum access and elections

T2 Quorum loss

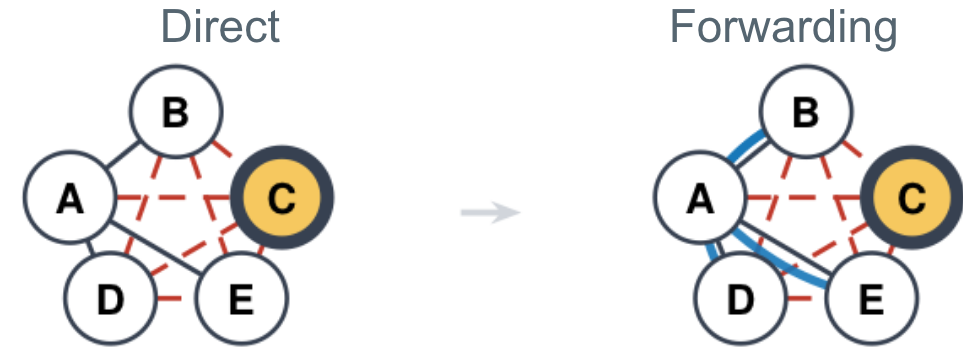


Leader C loses its direct majority.
A can relay traffic to other nodes.

Solid links survive. Dashed links fail. Blue paths forward traffic.

Controls: T0 healthy, T1 non-critical cut, T4 three-node chain

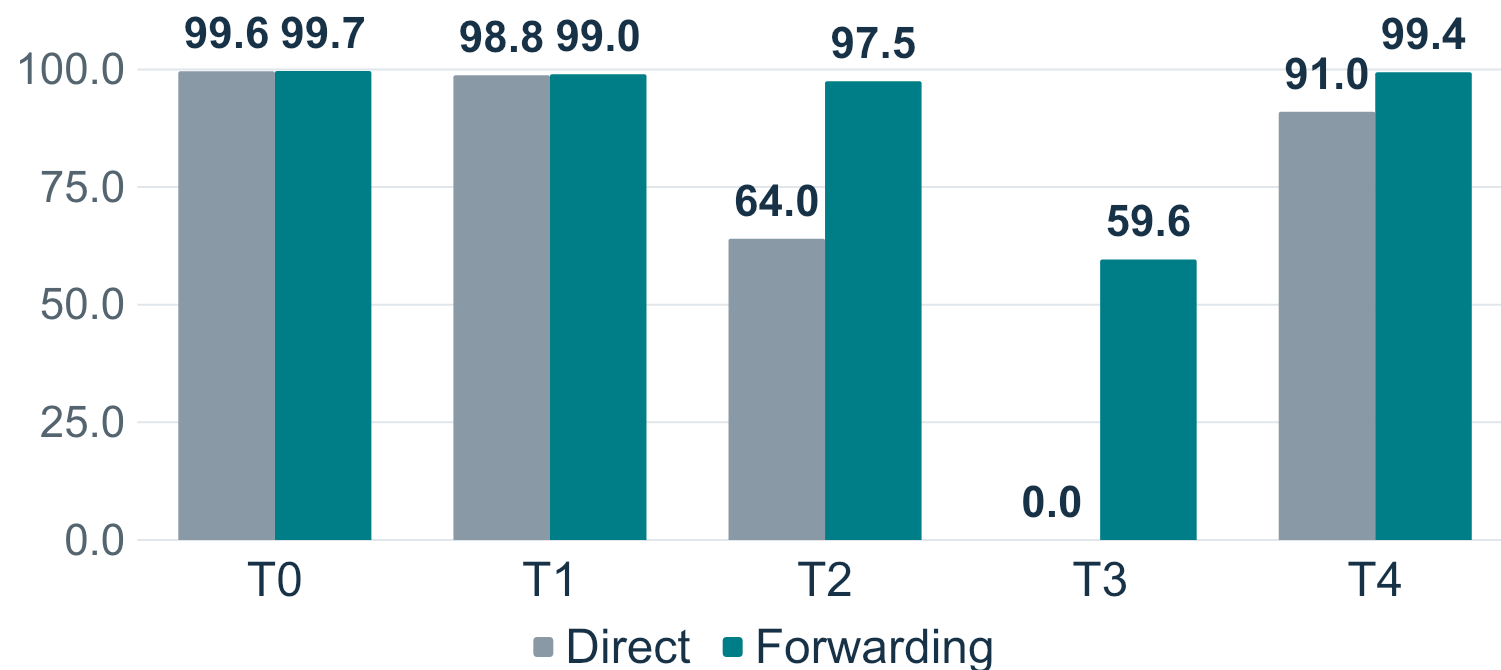
T3 Constrained election



C is isolated. A has an older log.
Forwarding lets B reach a majority.

Forwarding restores progress in T3

Commit throughput (entries/s)



T2 Quorum loss

+52%

higher mean throughput

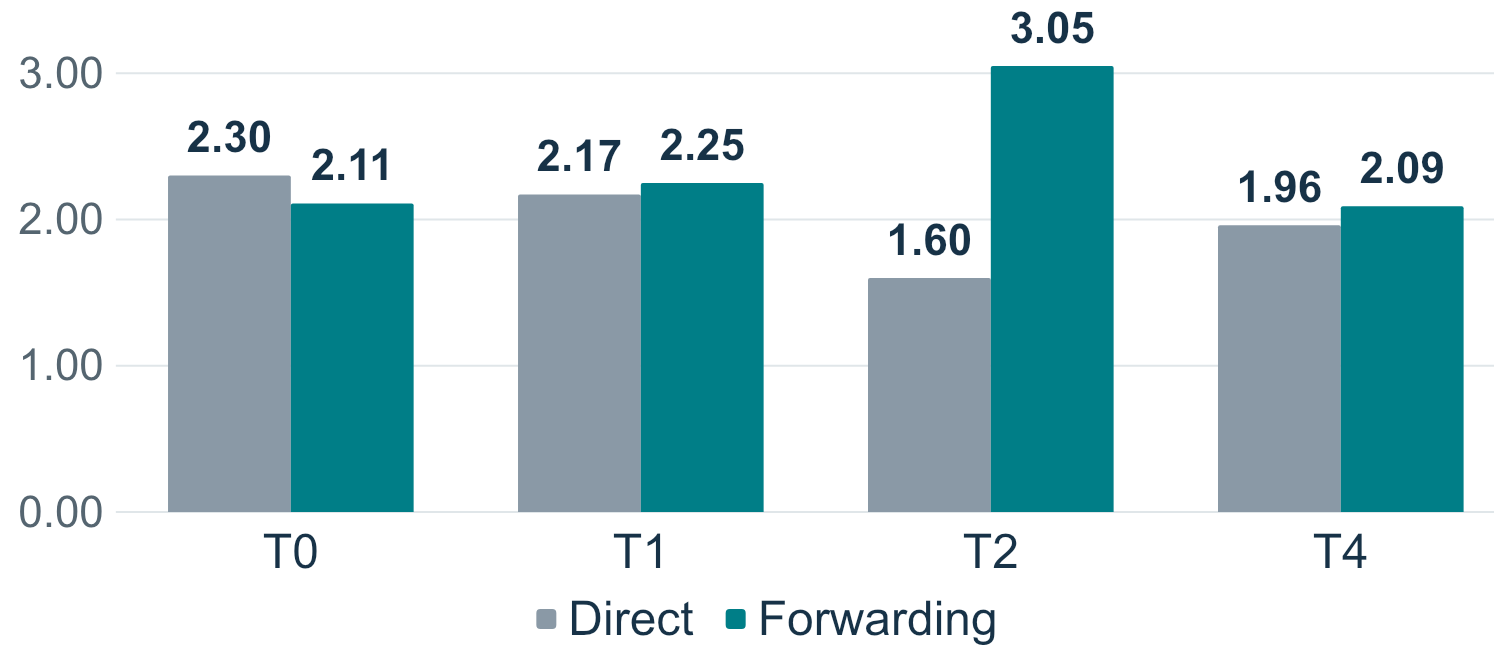
T3 Recovery

30/30 runs recover
20.27 s mean recovery

0.5 ms per link, 30 runs per condition. Bars show means.

A forwarded quorum adds commit latency

Mean of per-run median commit latency (ms)



T2 Quorum loss

1.60 to 3.05 ms

About twice the latency

T3 forwarding: 2.55 ms
No direct commits

0.5 ms per link, 30 runs per condition. Successful commits only.

Routing can improve resilience at a cost

Progress

Forwarding rescues T3 and improves T2 throughput.

Cost

Commits take longer when the majority needs extra hops.

Evidence focuses on 0.5 ms links.

At 5 and 20 ms, the fixed 10 ms heartbeat limits stable operation.

Future work

01 Timeouts matched to path delay

Repeat higher-delay trials with suitable or adaptive timeouts.

02 PreVote comparison

Compare routing with PreVote enabled and disabled.

03 Deployment across machines and regions

Test variable delays in a more realistic network.

Questions

Transparent routing for Raft under partial partitions